Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/24/2005
PI Name:	Sangani, Ashok S. Ph.D.		
Project Title:	Behavior of rapidly sheared bubble suspensions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Syracuse	**State:**	NY
Zip Code:	13244-1190	**Congressional District:**	25
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	09/01/1996	**End Date:**	08/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Koch, Donald (Cornell University)		
Grant/Contract No.:	NAG3-2427		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The objective of this experiment is to test the equations of motion of bubble suspensions in an ideal setting where potential flow prevails, and thereby validate the concepts of bubble pressure and effective viscosity of bubble suspensions. The conditions that promote potential flow are that the Reynolds number be large and the Weber number be small. The means to achieve the objectives are to generate a bubble suspension in a Couette cell in the microgravity environment and to shear it by rotating the outer cylinder.The forces acting on the bubbles in this configuration are the centrifugal force which pushes the bubbles toward the inner cylinder and the shear induced bubble phase pressure which tends to push the bubbles apart from each other. To accomplish the objective, measurements of the volume fraction, and the liquid and bubble velocity will be taken as functions of radial position. In addition the shear stress at the wall will be measured. Ground-based research objectives are: (i) to produce nearly monodisperse suspensions of nearly spherical, non-coalescing bubbles; (ii) to develop probes for measuring volume fraction and velocity distributions of the bubbles as a function of position; (iii) to measure wall shear stress; (iv) to perform experiments for bubble suspensions in vertical		

	and inclined channels and in circular, vertical pipe configurations; and (v) to continue analytical work related to the description of flows of bubble suspensions. Multphase flows occur in many industrial processes. The questions concerning the stability of well-dispersed multiphase medium and its properties are important in predicting the behavior of multiphase flows and in designing equipment in which these flows occur. The detailed investigation of the idealized bubble suspensions under consideration is a first step toward developing an analytical framework for answering these questions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bubbly liquids are encountered in nature and in many industrial processes. These processes can be designed better if the properties of bubbly liquids and their equations of motion are known. Understanding forces responsible for seggregation and clustering and coalescence is crucial for designing reliable equipment, e.g. bubble reactors and gas-liquid transport lines, that handle gas-liquid flows.
Task Progress:	PROGRESS REPORT Construction of a flow loop for carrying out bubbly flow experiments in a vertical pipe was completed and the probe for measuring wall stress was calibrated. Measurements were made for bubble phase volume fraction, velocity, radius, and velocity variance across the pipe cross-section along with the wall stress, and the liquid velocity variance for flows at various Reynolds numbers based on liquid flow rate and overall gas volume fraction. A manuscript based on this work is in preparation. Boundary conditions for bubbly flow near wall were derived based on a model of bubble-wall interactions and these were verified by comparing predictions based on the conditions with those obtained from detailed numerical simulations of a model of bubbly liquid flow. These boundary conditions together with the average equations of motion of bubbly liquids were solved using a spectral method to determine the profiles bubble phase volume fraction, velocity and variance in pipe flows. The predictions were found to be in a reasonably good agreement with the measured values. A manuscript based on this will be prepared in the near future.
Bibliography Type:	Description: (Last Updated: 04/02/2007)
Articles in Peer-reviewed Journals	Zenit, R.; Tsang, Y. H.; Koch, D. L.; and Sangani, A. S. "J. Fluid Mech." vol 515, pp. 261-292 , Oct-2004